HUAWEI HCCDA-AI

Linear Regression

Simple and Multiple Linear Regression



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Learning outcomes

LESSON GOALS

By the end of this lesson, students should be able to:

- Explain regression as the prediction of a number.
- Identify input features X and target values y.
- Read the equation $\hat{y} = wx + b$.
- Explain error, loss, derivative, and gradient descent.
- Distinguish simple regression from multiple regression.
- Train and evaluate a model with scikit-learn.



One connected story: examples → errors → learning → prediction

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A house-price question

REAL-WORLD PROBLEM

Can a model estimate the price of this house?

The model receives facts about the house. During training, it also sees the correct selling price.

Class question: Which house details may change its price?

Information given to the model

Area	**2,000 sq ft**
Bedrooms	**3**
Bathrooms	**2**
Stories	**2**
Parking spaces	**1**

Predicted price: ?

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Regression predicts a number

CORE IDEA



Input features

Size, bedrooms, bathrooms

Learned model

A mathematical relationship

Number prediction

Estimated house price

Regression learns from examples where both X and y are already known.

Regression output: 250,000

Classification output: expensive

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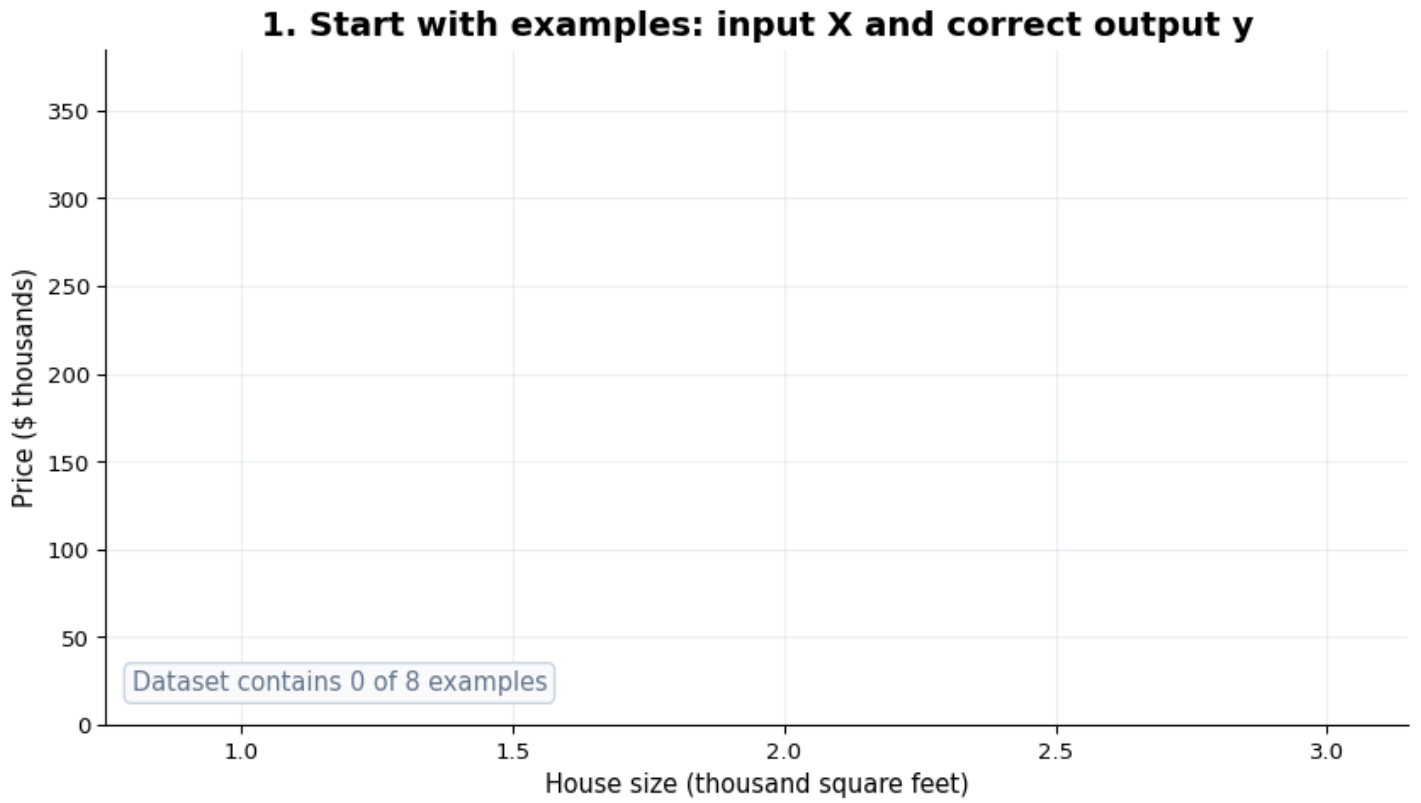
Training data: X and y

ANIMATED EXAMPLE

Before the model can learn, we give it examples.

- X is the house size.
- y is the known price.
- Each dot represents one house.
- More examples reveal the pattern.

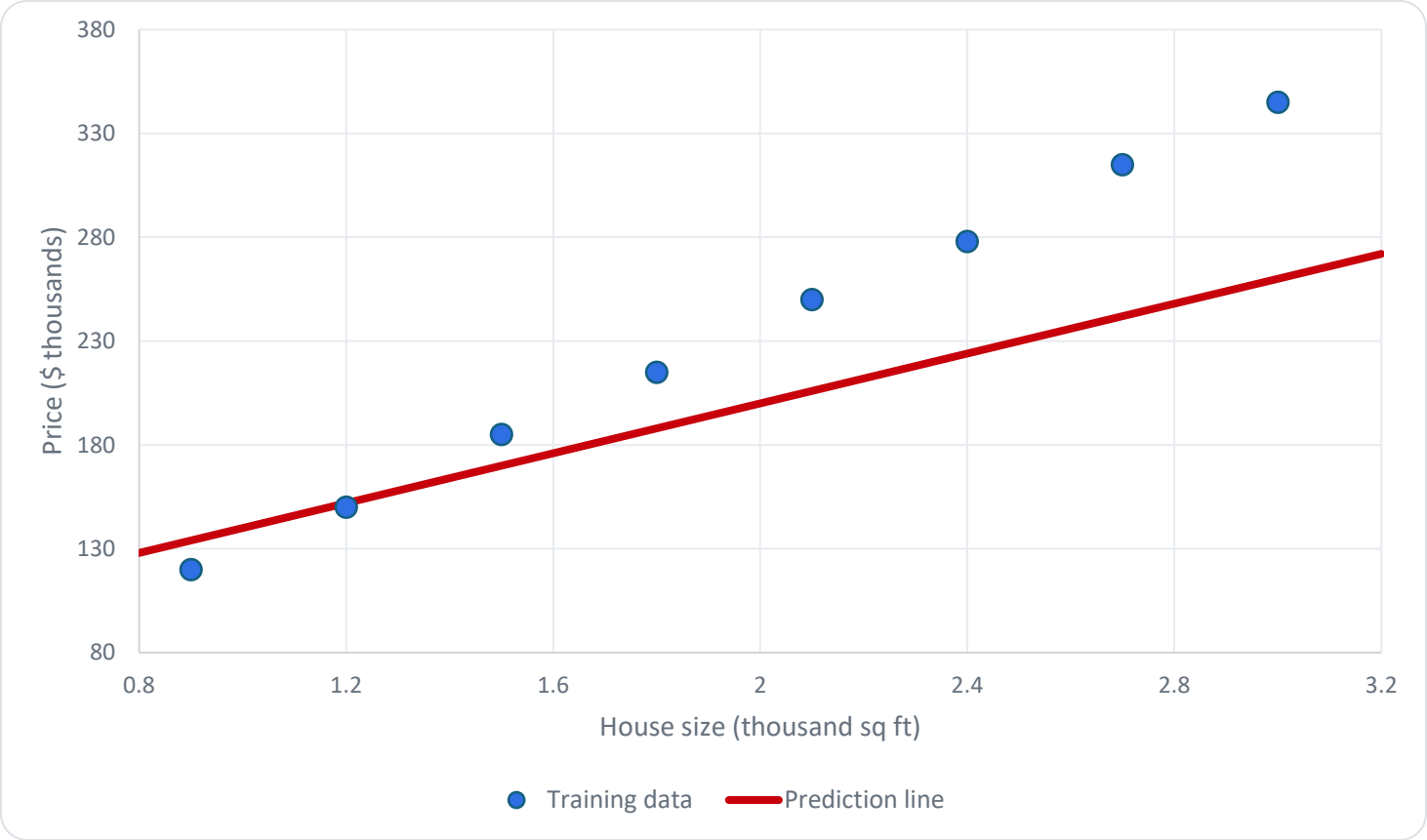
No learning yet. We are only showing the examples.



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A straight-line model

SIMPLE LINEAR REGRESSION



$$\hat{y} = wx + b$$

- $\hat{y}$ predicted price
- x input feature
- w slope of the line
- b starting position

Simple regression uses one input feature.

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Prediction error

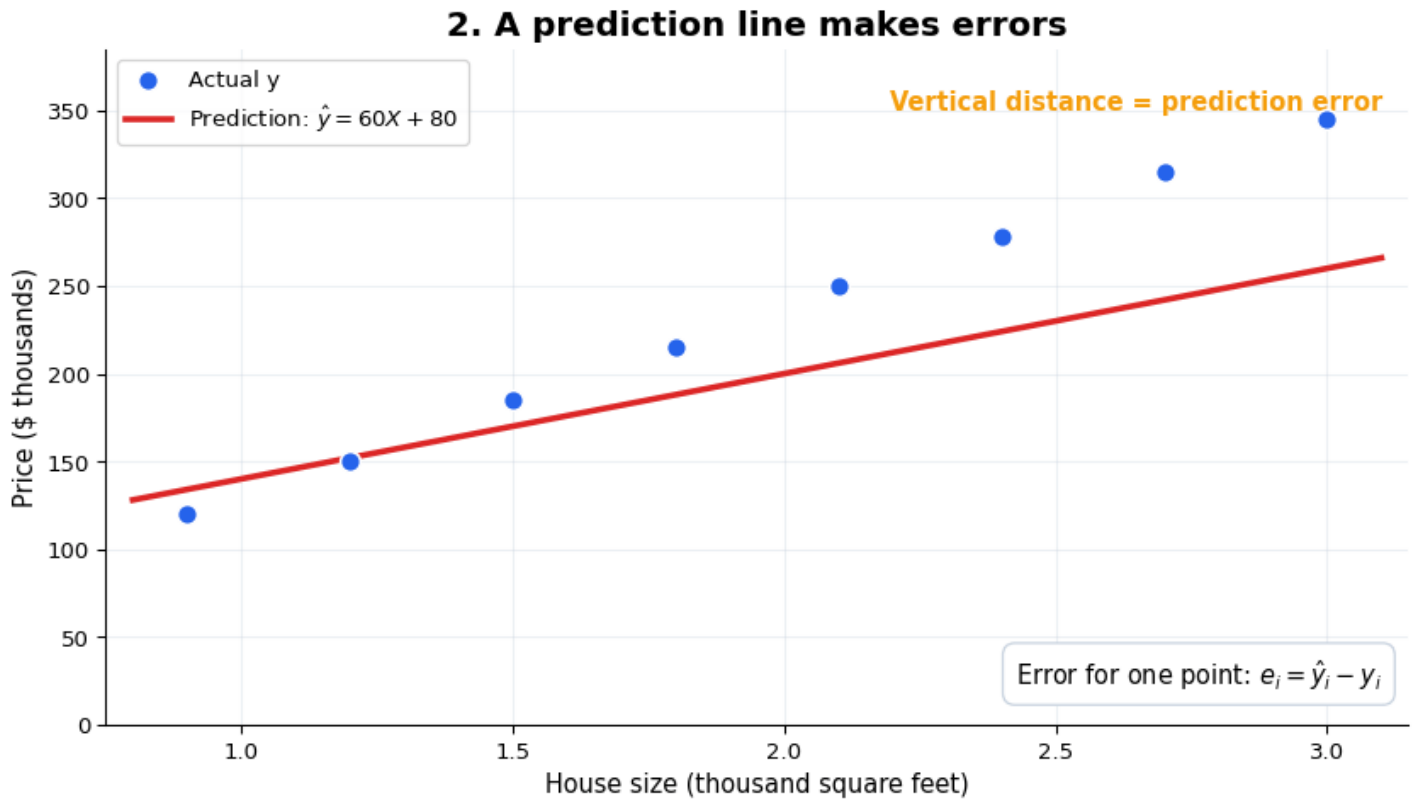
MEASURING ONE PREDICTION

error = prediction − actual

Example

Prediction: 200
Actual: 250
Error: 200 − 250 = −50

- Zero means an exact prediction.
- A larger distance means a larger mistake.



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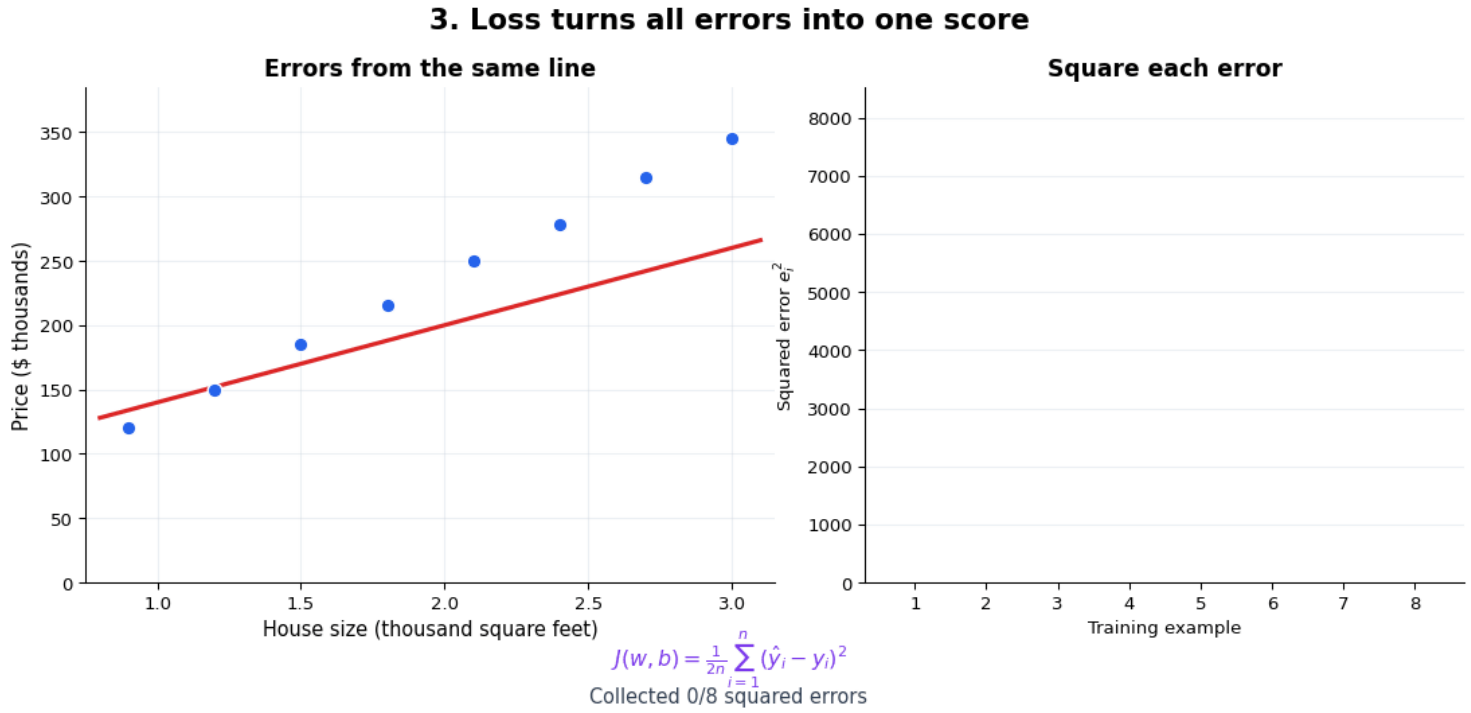
Loss: one score for all errors

MEASURING THE WHOLE LINE

1. Calculate every error
2. Square each error
3. Add and average them

$$J(w,b) = 1/(2n)\ \Sigma(\hat{y} - y)^2$$

Smaller loss means the line fits the data better.

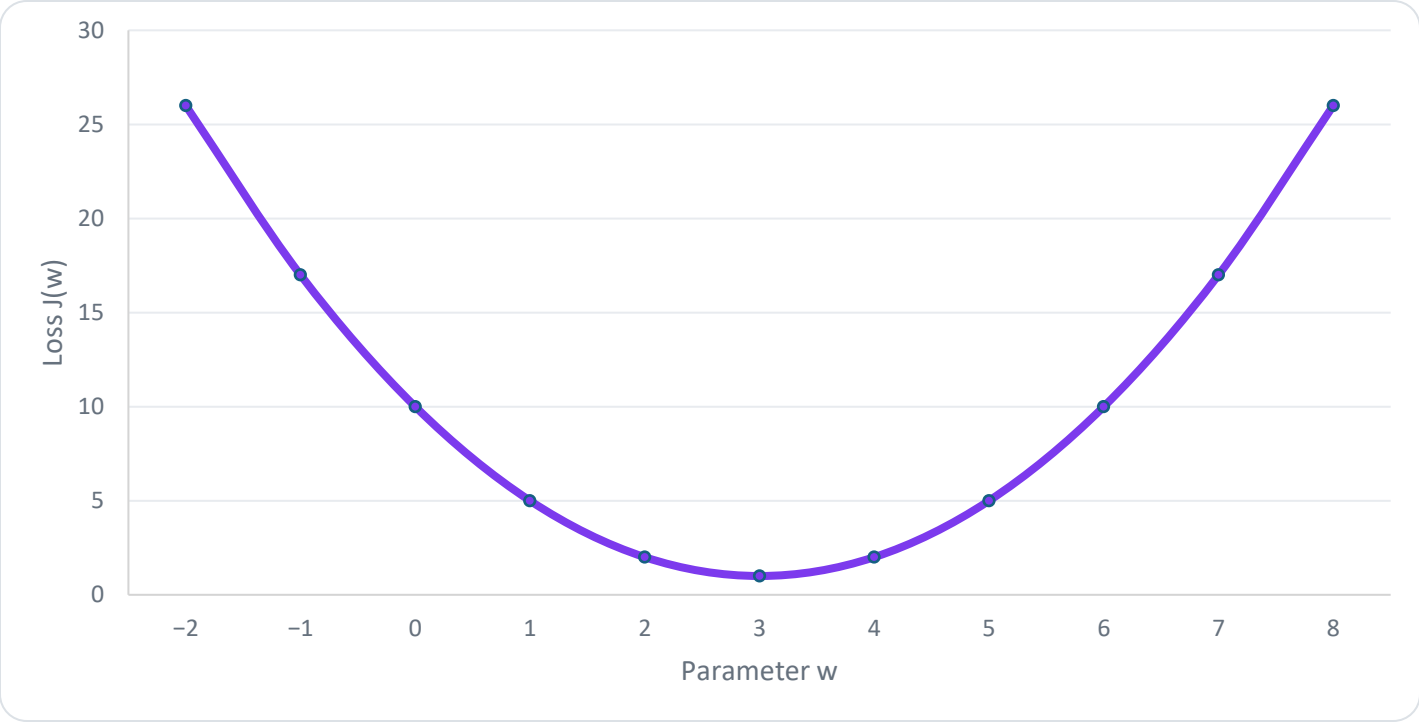


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Derivative: direction and steepness

WHY THE MODEL KNOWS WHERE TO MOVE

A derivative tells how the loss changes when a parameter changes slightly.



$$dJ/dw = 2(w - 3)$$

Negative derivative

Increasing w lowers the loss.

Zero derivative

The curve is flat at the minimum.

Positive derivative

Increasing w raises the loss.

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Gradient descent

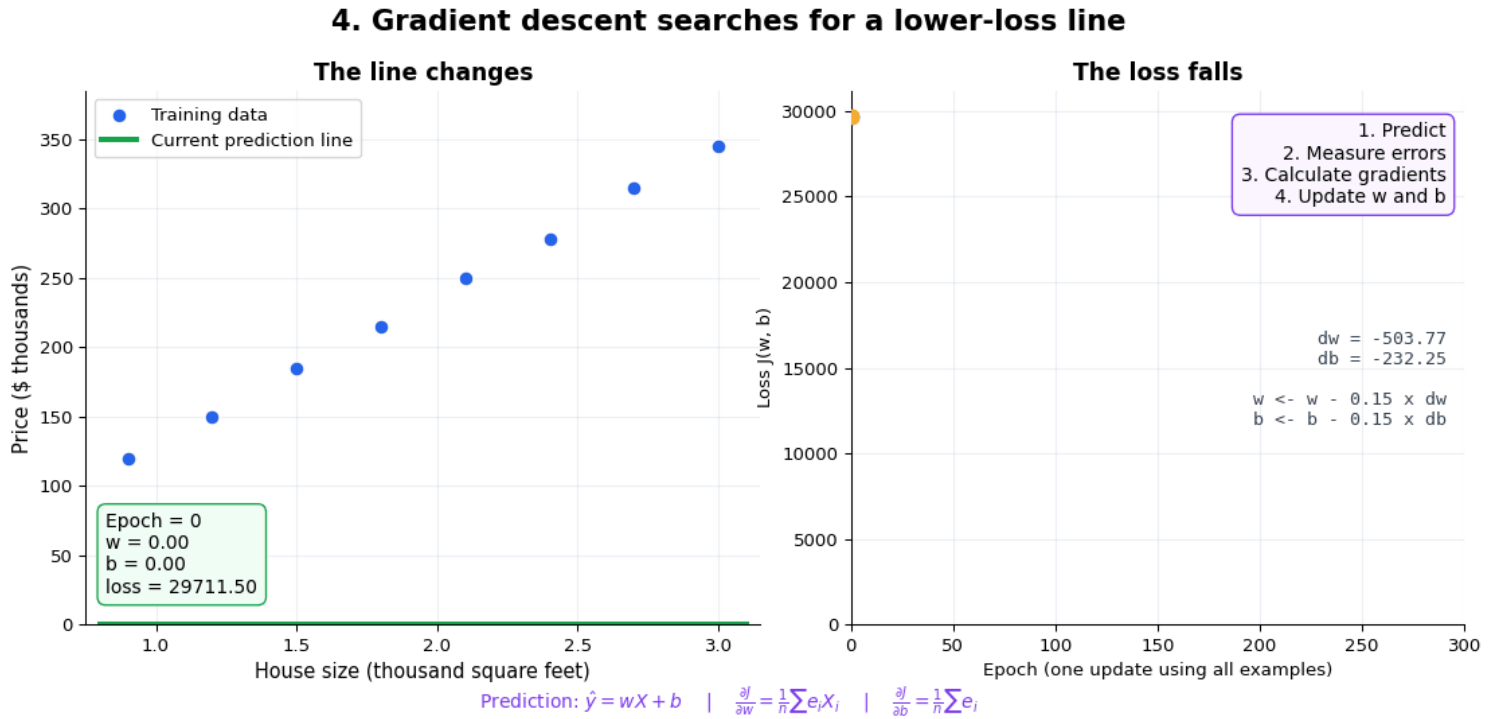
USING THE DERIVATIVE TO LEARN

Gradient descent repeatedly changes w and b to reduce the loss.

$$w = w - \alpha \cdot dw$$
$$b = b - \alpha \cdot db$$

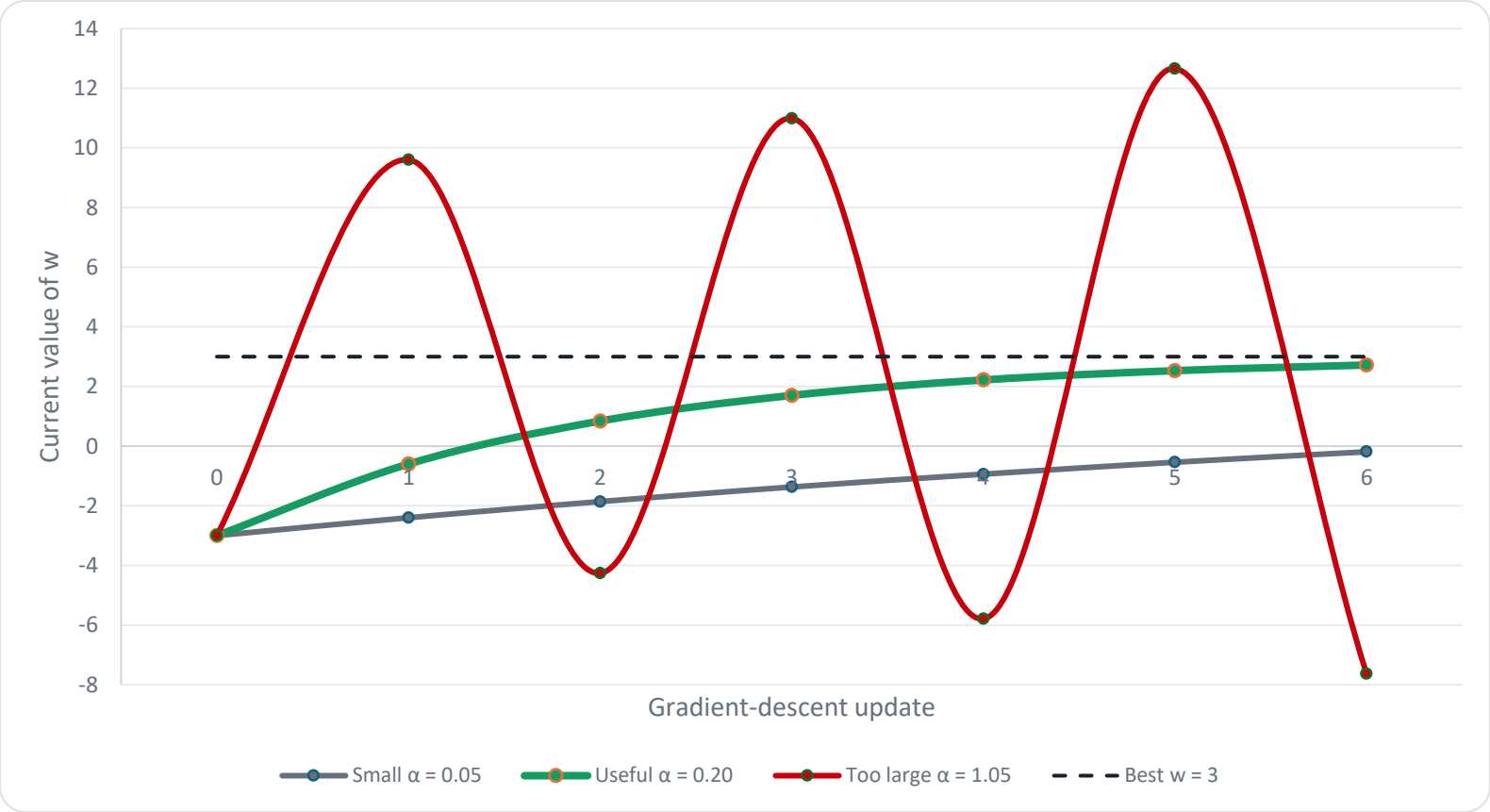
α is the learning rate.
dw and db are the gradients.

We subtract because the derivative points uphill.



Learning rate controls step size

CHOOSING HOW FAR TO MOVE



Small

Safe, but learning is slow.

Useful

Moves steadily toward the minimum.

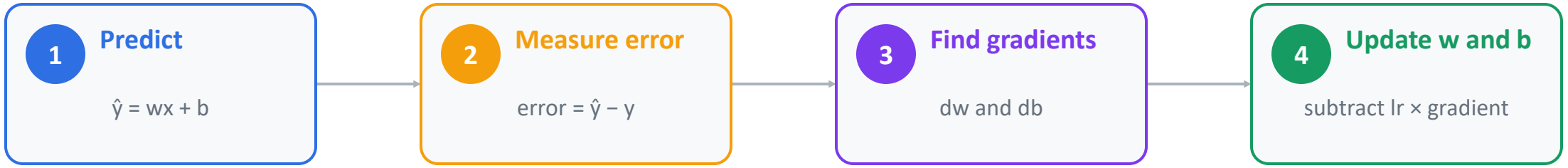
Too large

Overshoots and may move farther away.

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The training loop

FROM MATHEMATICS TO PYTHON




```
for epoch in range(epochs):
    y_hat = w * X + b
    error = y_hat - y
    dw = mean(error * X)
    db = mean(error)
    w = w - lr * dw
    b = b - lr * db
```

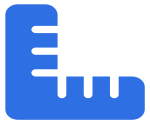
The loop repeats until the loss stops improving or the chosen number of epochs finishes.

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Multiple linear regression

MORE THAN ONE INPUT FEATURE

A house price may depend on several facts at the same time.



Area

2,000 sq ft



Bedrooms

3



Bathrooms

2



Stories

2



Parking

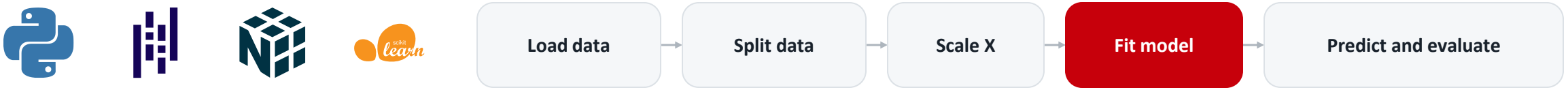
1

$$\hat{y} = b + w_1(\text{area}) + w_2(\text{bedrooms}) + w_3(\text{bathrooms}) + w_4(\text{stories}) + w_5(\text{parking})$$

One weight is learned for each feature. The model combines their effects to make one price prediction.

Scikit-learn workflow

TRAINING THE MODEL




```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

model = LinearRegression()
model.fit(X_train_scaled, y_train)
y_pred = model.predict(X_test_scaled)
```

Important

Fit the scaler only on the training data.

Use the same scaler for test data and new houses.

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Evaluation and key ideas

CHECKING THE MODEL

Metric	What it tells us	Better value
MSE	Average squared prediction error	Lower
R^2	Improvement over predicting the average	Closer to 1

$R^2 = 0.55$

The model explains about 55% of the price variation in the test data.

This does not mean 55% accuracy.

Remember

- Simple linear regression uses one input feature.
- Multiple linear regression combines several features.
- Loss measures how wrong the predictions are.
- Derivatives guide gradient-descent updates.
- Test data checks performance on unseen examples.

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